



Research Intake 2026

Day 37

Explainable AI & Medical Imaging

A Beginner's Guide for Students

Learning Computer Vision Through Images, Videos, and Artificial Intelligence

Gudsky Research Foundation

Table of Contents

1 Why Explainability Matters in Vision	3
2 Interpretability vs. Explainability	4
3 Gradient-Based Explanation Methods	6
3.1 Grad-CAM Worked Walkthrough.....	7
4 Perturbation-Based Explanation Methods	8
5 Attention-Based Explanations.....	11
6 Medical Imaging as a High-Stakes CV Domain	13
7 Medical Image Modalities & Data Characteristics	14
8 Architectures for Medical Image Analysis	16
9 Evaluation, Regulation & Clinical Deployment	18
10 Module Synthesis & Research Frontiers.....	20
11 Common Pitfalls: A Practical Debugging Checklist.....	23
Glossary of Terms.....	24
Chapter Summary: Key Takeaways.....	25
Assignment / Practical Project Brief.....	26
A Core Requirements	26
B Dataset	27
C Deliverables.....	27
D Grading Rubric	27
E Tips and Common Pitfalls	28
F Submission Checklist.....	28

The future of AI is bright, and you can be part of it!

1 Why Explainability Matters in Vision

This day is where two threads that have run separately through this entire compendium finally converge. The first thread is interpretability itself: Day 29 §9 introduced filter visualization, feature map inspection, and Grad-CAM as tools for understanding what a trained CNN had actually learned, and flagged explicitly that Grad-CAM's role would become "centrally important, not just a nice-to-have" once this day's medical imaging content arrived. The second thread is medical imaging as an application domain: previewed in Day 27 §9.1's modality-specific preprocessing, Day 33 §7.3's Dice coefficient discussion, and Day 36 §6's volumetric representation content, but never yet treated as a domain in its own right. This day treats both fully, and shows why they belong together rather than as two separate closing topics.

The connection is not incidental. Day 26 §9.2 introduced the Gender Shades study and the broader concern that a model's high aggregate accuracy can mask serious, unevenly distributed failure — a concern this day extends to its highest-stakes form. A wrong classification or a wrong detection in most of this compendium's earlier applications (Day 31's retail inventory monitoring, Day 32's sports analytics) produces an inconvenience or a minor cost. A wrong classification in a diagnostic imaging context can directly affect a patient's treatment and outcome. This asymmetry is why medical imaging, more than any other domain this compendium has covered, cannot tolerate a model whose predictions are simply trusted without explanation.

1.1 The Specific Risk of a Confident, Plausible-Looking Wrong Answer

It is worth being precise about why "black box" predictions are especially risky in vision specifically, as opposed to other machine learning applications. A vision model's wrong prediction is not obviously wrong the way a nonsensical text output might be — a misclassified chest X-ray still looks, to a human reviewer without independent expertise, like a perfectly ordinary chest X-ray with a label attached. The output carries no visible signal of its own unreliability. A confidently wrong prediction is, in this sense, more dangerous than an uncertain one: it invites exactly the kind of unquestioning trust that produces a harmful downstream decision.

Callout Box: *The wolves-versus-huskies case, previewed already in Day 29 §9.3's discussion of Grad-CAM's debugging value, is worth restating in full here as this day's motivating example. A classifier trained to distinguish wolves from huskies achieved excellent held-out accuracy — and, on inspection with exactly the gradient-based methods Section 3 develops, turned out to have learned almost nothing about the animals themselves. It had learned that wolf photographs tend to have snowy backgrounds and husky photographs tend not to. The model was confidently, plausibly, and completely wrong about what it was actually detecting — and no amount of inspecting its accuracy figures alone would ever have revealed this. Only a method that could show which pixels the model's decision actually depended on made the failure visible.*

1.2 Where This Day Sits in the Module's Arc

This is the final day of the Computer Vision Fundamentals module, and it is worth previewing explicitly how deliberately it draws on nearly everything that came before it, since Section 10 develops this synthesis in full. Medical image classification (Section 8) exercises the CNN transfer-learning content from Day 29 §7. Medical image segmentation (Section 8) is the direct payoff of Day 33 §4's U-Net forward reference. Volumetric CT and MRI data (Section 7) connects directly to Day 36 §6's point cloud, voxel grid, and representation-tradeoff content. And Sections 3 through 5's explainability methods are the deferred technical development of a capability introduced piecemeal since Day 29 §9, now finally given the full, dedicated treatment its eventual application in Section 6 onward requires.

1.3 This Day's Practical Project

This day carries a graded assignment, PP13, developed in full at the end of this document. PP13 is deliberately structured as a two-part capstone exercising both of this day's major architectural threads: a 2D classification task (Task A, echoing Day 26's PP12) and a segmentation task (Task B, echoing Day 33's U-Net content), with a required explainability component attached to both — making Sections 3 through 5 and Sections 7 through 8 all directly load-bearing for the assignment, not background reading.

2 Interpretability vs. Explainability

These two terms are frequently used loosely and interchangeably in casual discussion, but the distinction between them is worth drawing precisely before developing any specific method, since it determines which category every technique in Sections 3 through 5 actually belongs to.

2.1 A Definitional Distinction

Interpretability refers to a model whose internal structure is transparent enough that a human can understand its reasoning directly, by inspecting the model itself — no separate explanation-generating step is required, because the structure is the explanation. Explainability refers to a technique that produces a post-hoc justification for an otherwise opaque model's specific decision, without making the model's internal computation itself any more transparent. A linear regression model is interpretable: its prediction is a weighted sum of input features, and the weights themselves directly show which features mattered and by how much, in a form a human can read off immediately. A deep CNN is not interpretable in this sense — millions of parameters spread across dozens of non-linear layers offer no comparably direct human-readable account of why a specific prediction was made — which is exactly why Sections 3 through 5 are all explainability techniques rather than interpretability techniques.

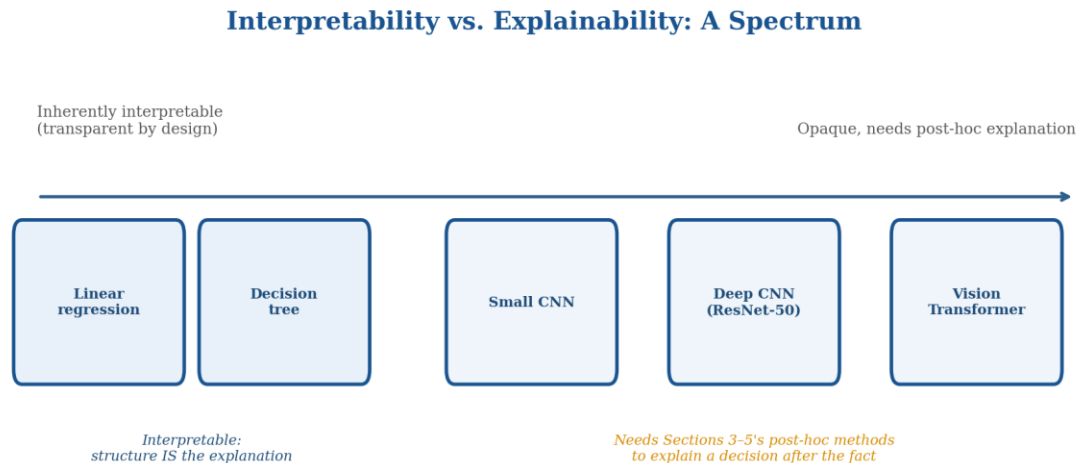


Figure 1. Interpretability and explainability sit at opposite ends of a spectrum, not as a strict binary. Simple models (left) are transparent by design — their structure is the explanation. Modern CNNs and Vision Transformers (right) are not, and require the post-hoc explanation methods developed in Sections 3 through 5 to make their decisions inspectable at all.

2.2 Where CNNs and Vision Transformers Fall on This Spectrum

Every architecture this compendium has covered since Day 29 sits toward the opaque end of Figure 1's spectrum. A CNN's convolutional filters (Day 29 §9.1) can be visualized directly, and early-layer filters are often genuinely interpretable on inspection — they resemble simple edge or color-blob detectors, as Day 29 §9.1 showed directly. But this interpretability degrades sharply with depth: a filter several layers deep operates on already-abstracted feature maps, not raw image patches, and no longer corresponds to any humanly recognizable visual pattern on its own. A Vision Transformer's self-attention weights (previewed conceptually in Day 30 §7, developed in this compendium's later Vision Transformers module) are sometimes presented as more inherently interpretable than a CNN's convolutional filters, since attention weights directly indicate which input patches a given layer weighted most heavily — but Section 5.2 develops an important caveat to this claim directly.

2.3 Why Most of This Day's Content Is Post-Hoc Explanation

Given that neither CNNs nor Vision Transformers are inherently interpretable in the strict sense Section 2.1 defined, the practical question this day actually needs to answer is not "how do we make the model transparent" but "how do we produce a trustworthy after-the-fact account of what a given opaque model actually did." This is precisely what Sections 3 through 5 develop: three distinct families of post-hoc explanation technique, differing in what information they require access to (gradients, for Section 3; only input-output behavior, for Section 4; attention weights specifically, for Section 5) and in how faithfully their explanation reflects the model's actual internal computation — a question worth holding onto

explicitly, since not every explanation method's output is equally trustworthy, a caveat Section 5.2 in particular treats with real care rather than glossing over.

Research Spotlight: Is a Post-Hoc Explanation Always Faithful?

A subtle but important caveat applies to every technique in Sections 3 through 5: a post-hoc explanation method is, itself, a separate computational procedure layered on top of the original model — and there is no guarantee, in general, that its output faithfully reflects the reasoning the original model actually performed, rather than merely producing a plausible-looking but potentially misleading account after the fact.

This is not a reason to dismiss these methods; the wolves-versus-huskies case in Section 1.1 is direct evidence that gradient-based explanation can reveal a genuine, otherwise-invisible failure. It is a reason to treat any single explanation's output with the same epistemic caution this compendium has applied throughout — a Grad-CAM heatmap or an attention map is evidence to be weighed, not a certified, ground-truth account of the model's reasoning.

3 Gradient-Based Explanation Methods

This section develops in full the technique Day 29 §9.3 introduced and explicitly deferred: Grad-CAM, together with the broader family of gradient-based explanation methods it belongs to. All of these methods share one defining property: they use the gradient of a model's output with respect to its input (or an intermediate representation) as the raw signal an explanation is built from — information already available from a single backward pass, since backpropagation (Day 29 §7.1) computes exactly this gradient as a normal part of training.

Saliency Maps: The Simplest Gradient-Based Method

The most direct gradient-based explanation computes the gradient of a target class's predicted score with respect to every individual input pixel, producing a saliency map the same size as the input image: large-magnitude gradient values indicate pixels where a small change would most affect the prediction, and are interpreted as pixels the model's decision was most sensitive to. This is conceptually the simplest member of the family — no additional machinery beyond a single backward pass to the input layer — but in practice tends to produce visually noisy, scattered results that can be difficult to interpret directly, motivating Grad-CAM's more structured alternative.

Grad-CAM: Mechanics

Grad-CAM (Gradient-weighted Class Activation Mapping), recapped briefly in Day 29 §9.3, computes the gradient of a target class's predicted score not with respect to the raw input pixels, but with respect to a chosen convolutional layer's feature maps — typically the network's last convolutional layer, since it

combines the deepest, most class-discriminative features with still-meaningful spatial resolution. Figure 2 lays out the full pipeline. These gradients are globally averaged across each feature map's spatial dimensions to produce one importance weight per channel; each channel of the chosen layer's feature maps is then scaled by its corresponding importance weight and the weighted channels are summed together, producing a single coarse spatial map; a ReLU is applied to retain only the pixels that positively support the target class, discarding evidence for competing classes. The result is a low-resolution heatmap, upsampled back to the input image's size for visualization, highlighting the image regions that most positively contributed to the target class's predicted score.

Grad-CAM Mechanics: From Gradients to a Localization Heatmap

backward pass: one gradient computation, §3.1

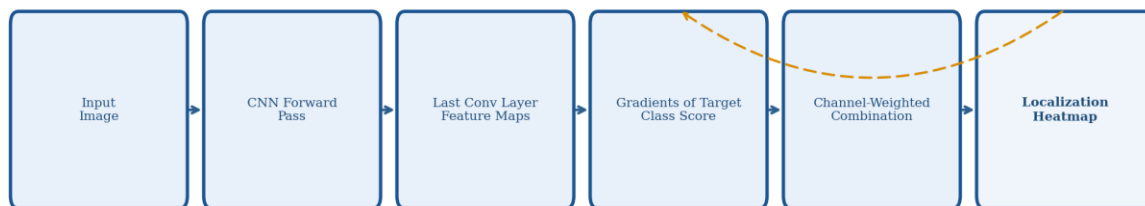


Figure 2. Grad-CAM's full pipeline: a single backward pass computes gradients of the target class score with respect to the last convolutional layer's feature maps; these gradients are globally averaged into per-channel importance weights, which combine the feature maps into a single coarse localization heatmap. The entire process requires one forward pass and one backward pass — no retraining, no additional model.

3.1 Grad-CAM Worked Walkthrough

Consider a ResNet-50 classifier, fine-tuned via exactly the transfer-learning workflow Day 29 §7.5 developed, to distinguish pneumonia from normal chest X-rays — precisely the PP13 Task A scenario this day's assignment develops in full. Given a specific X-ray the model classifies as "pneumonia" with high confidence, Grad-CAM is applied by: (1) running a forward pass to obtain the predicted pneumonia score; (2) computing the gradient of that score with respect to the final convolutional layer's feature maps via one backward pass; (3) globally averaging these gradients per channel to obtain importance weights; (4) combining the feature maps using these weights into a single heatmap; (5) overlaying the upsampled heatmap on the original X-ray. A clinically sensible result shows the heatmap concentrated over the specific lung region with visible opacity consistent with pneumonia — directly corroborating that the model's high confidence reflects genuine, clinically relevant evidence rather than a spurious correlation of exactly the kind Section 1.1's wolves-versus-huskies case warns against.

Worked Example: Why the Choice of Layer Matters

Grad-CAM's heatmap resolution is capped by the spatial resolution of whichever convolutional layer's feature maps are chosen — this is a direct consequence of Day 29 §4's pooling-driven downsampling: a network's last convolutional layer, having passed through several pooling stages, may have a spatial resolution as coarse as 7×7 for a 224×224 input image.

Choosing an earlier, higher-resolution layer produces a sharper-looking heatmap, but at the cost of using less class-discriminative features — early layers, per Day 29 §9.1, detect generic edges and textures rather than the class-specific patterns a later layer has learned to recognize.

This is a genuine tradeoff, not a bug to engineer away: Grad-CAM applied to the last convolutional layer (the standard default) trades spatial precision for semantic relevance, and this coarse-but-relevant heatmap is what §3.3 identifies as the technique's central limitation.

Grad-CAM++ and Further Refinements

Grad-CAM++ refines the original formulation's channel-weighting step with a more careful pixel-wise weighting scheme, producing measurably better localization specifically when an image contains multiple instances of the same class — a scenario where the original Grad-CAM formulation's simple global-average weighting can under-highlight some instances in favor of others. This is a targeted fix to a known weakness rather than a wholesale replacement, and the original Grad-CAM formulation, developed in full above, remains the standard default for the single-object classification scenario PP13's Task A presents.

Why Gradient-Based Methods Are Popular, and Their Limitations

Gradient-based methods are computationally cheap relative to the alternatives Section 4 develops — a single additional backward pass, no repeated model evaluation — which is a large part of why Grad-CAM in particular has become the default, first-choice explanation method across this compendium's applications. Their central limitation is the resolution-versus-relevance tradeoff Section 3.2's worked example just developed: the resulting heatmap is a coarse approximation, not a pixel-precise account of the model's reasoning, and should be read at the level of "which broad region mattered" rather than "which exact pixels mattered." Gradient-based methods can also be noisy and sensitive to small, sometimes visually imperceptible input perturbations — a saliency map computed for two nearly identical images can occasionally look surprisingly different, a fragility worth being aware of when interpreting any single result with high confidence.

4 Perturbation-Based Explanation Methods

Where Section 3's gradient-based methods require direct access to a model's internal gradients, this section develops a family of methods that require no such access at all — treating the model purely as a black box

that maps an input to an output, and explaining a prediction entirely by observing how that output changes as the input is deliberately, systematically modified.

4.1 Occlusion Sensitivity

Occlusion sensitivity is the most direct perturbation-based method: systematically slide a small mask (commonly a solid gray or black patch) across every position of an input image, re-running the model's forward pass at each position, and recording how much the target class's predicted confidence drops when that specific region is occluded. Figure 3 shows the resulting confidence-drop map: a sharp drop when a specific region is masked indicates that region was important to the original prediction; little to no change indicates the model's decision did not depend meaningfully on that region.

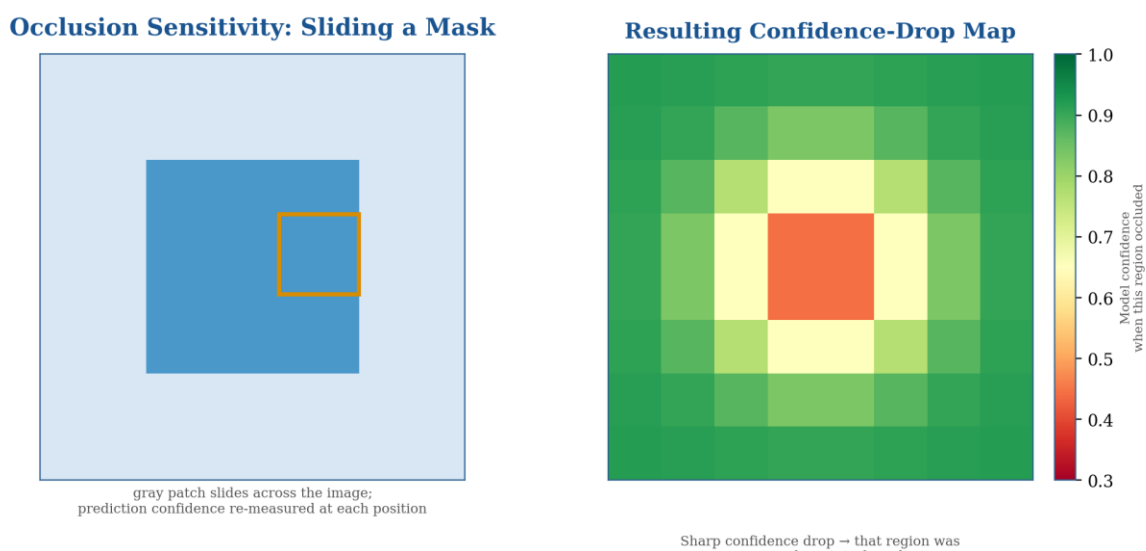


Figure 3. Occlusion sensitivity: a mask slides across the image (left), and the model's prediction confidence is re-measured with each region occluded in turn, producing a confidence-drop map (right). Sharp drops indicate regions the original prediction depended on — an explanation requiring no gradient access at all, only repeated forward passes.

4.2 LIME: Superpixel-Based Local Surrogate Models

LIME (Local Interpretable Model-agnostic Explanations) takes a related but more structured approach: rather than sliding a fixed-shape mask, the image is first segmented into superpixels — small, visually coherent regions, conceptually related to (though computed differently from) the segmentation output Day 33 developed at length — and a large number of perturbed versions of the image are generated by randomly turning different subsets of superpixels on or off. The model's predictions on these many perturbed versions are then used to fit a simple, inherently interpretable surrogate model (commonly a sparse linear model, exactly the kind of transparent-by-design model Section 2.1 described) that approximates the original opaque model's behavior locally, in the immediate neighborhood of the specific prediction being explained. The surrogate model's coefficients — directly interpretable, since it is a simple linear model — then indicate which superpixels contributed positively or negatively to the original prediction.

4.3 SHAP Values Applied to Vision

SHAP (SHapley Additive exPlanations) applies a concept from cooperative game theory — the Shapley value, which fairly attributes a total payoff among multiple contributing players — to the problem of attributing a model's prediction among its input features (superpixels or individual pixels, in the vision context). Each feature's SHAP value reflects its average marginal contribution to the prediction, computed across every possible combination of other features being present or absent — a more theoretically principled attribution than LIME's local linear approximation, at meaningfully higher computational cost, since exact Shapley value computation scales poorly with the number of features being attributed and is typically approximated in practice for anything beyond a small feature set.

4.4 Tradeoffs Versus Gradient-Based Methods

Property	Gradient-Based (§3)	Perturbation-Based (§4)
Requires model internals?	Yes — gradient access needed	No — treats model as a black box
Computational cost	Low — one backward pass	Higher — many forward passes
Works on any model type?	Only differentiable models	Yes — fully model-agnostic
Typical use case	Fast, default first check (§3.4)	Model-agnostic settings; deeper verification

The practical choice between these two families echoes the same "no universal winner, only a fit for constraints" pattern this compendium has established repeatedly (Day 30 §9, Day 31 §8, Day 34 §9.5) — gradient-based methods for a fast, cheap, first-pass explanation on a differentiable model; perturbation-based methods when gradient access is unavailable, when a model-agnostic explanation is specifically required, or when a gradient-based method's result needs independent verification through a structurally different technique.

Real-World Use Case: When Model-Agnostic Explanation Matters

Model-agnosticism is not merely a theoretical property — it has direct, practical value whenever an explanation needs to be produced for a model whose internals are inaccessible, such as a proprietary third-party diagnostic system integrated into a hospital's imaging workflow via an API that returns only a prediction, with no gradient access at all.

In this setting, Section 3's entire gradient-based family is simply unusable, and Section 4's perturbation-based methods are the only available option for producing any explanation at all — a genuine, non-hypothetical scenario in real clinical deployment, where a hospital adopting a vendor's diagnostic model may have no access to its internal architecture or weights.

5 Attention-Based Explanations

Day 30 §7 introduced Vision Transformers as an architectural alternative to convolution, built around self-attention rather than local receptive fields. This section develops a direct consequence of that architectural choice: a ViT's attention weights are, at least superficially, a built-in, ready-made explanation signal — every layer directly computes which input patches it weighted most heavily, with no additional gradient computation or perturbation procedure required at all.

5.1 Attention Maps as a Built-In Explanation Signal

For a given query patch (most commonly the classification token, per this compendium's later Vision Transformers module), a self-attention layer computes a weight for every other patch in the image, indicating how much that patch's information contributed to the query's updated representation at this layer. Visualized directly as a heatmap over the input image, these attention weights show, layer by layer, which regions the model was "looking at" — a genuinely different and, on the surface, more direct signal than Section 3's gradient-based reconstruction or Section 4's perturbation-based inference, since attention weights are not a post-hoc reconstruction at all but a value the model computed as part of its normal forward pass.

5.2 The Faithfulness Caveat: An Active Research Debate

It is important to be honest about a real limitation here rather than presenting attention maps as a strictly superior alternative to Sections 3 and 4's methods. A growing body of research has raised a specific, carefully-argued concern: high attention weight on a given patch does not necessarily mean that patch was actually important to the model's final decision, and conversely, a patch the model's decision depended heavily on does not necessarily receive high attention weight at every layer. Attention is one signal the model computes internally, but it is not obviously the same thing as "which input regions actually drove this prediction" — the same faithfulness caveat Section 2.3's callout raised in the abstract for every post-hoc explanation method, now shown to apply even to a signal that looks, at first glance, like a direct rather than reconstructed explanation.

Research Spotlight: Does Attention Explain, or Merely Correlate?

This debate is worth engaging with directly rather than resolving artificially in either direction. Proponents of attention-based explanation point out that attention weights are computed directly by the model as part of its actual forward pass, unlike Section 3's gradient reconstruction or Section 4's perturbation-based inference, both of which are explicitly separate procedures layered on top of the model after the fact.

Critics point to controlled experiments showing that a model's predictions can remain largely unchanged even when its attention weights are deliberately altered or replaced with alternative, equally plausible-looking attention patterns — evidence that attention weights alone may not uniquely determine, or fully account for, the model's actual decision.

The honest, current state of this research question is unsettled rather than resolved, and worth treating exactly that way: attention maps are a genuinely useful, cheap, readily-available signal worth inspecting, but — per Section 2.3's general caution — not a certified, ground-truth account of model reasoning, any more than a Grad-CAM heatmap is.

5.3 Attention Rollout: Aggregating Across Layers

A further complication: a Vision Transformer has many layers, each computing its own separate attention pattern, and no single layer's attention map alone gives a complete account of how information actually flowed from the input all the way to the final prediction. Attention rollout addresses this by combining attention weights across all layers — accounting for the fact that a patch's influence in a later layer depends on how information already flowed through it in every earlier layer — producing one aggregated attention map that better reflects the network's full depth, rather than an arbitrarily-chosen single layer's necessarily partial view. Figure 4 shows this aggregation process directly.

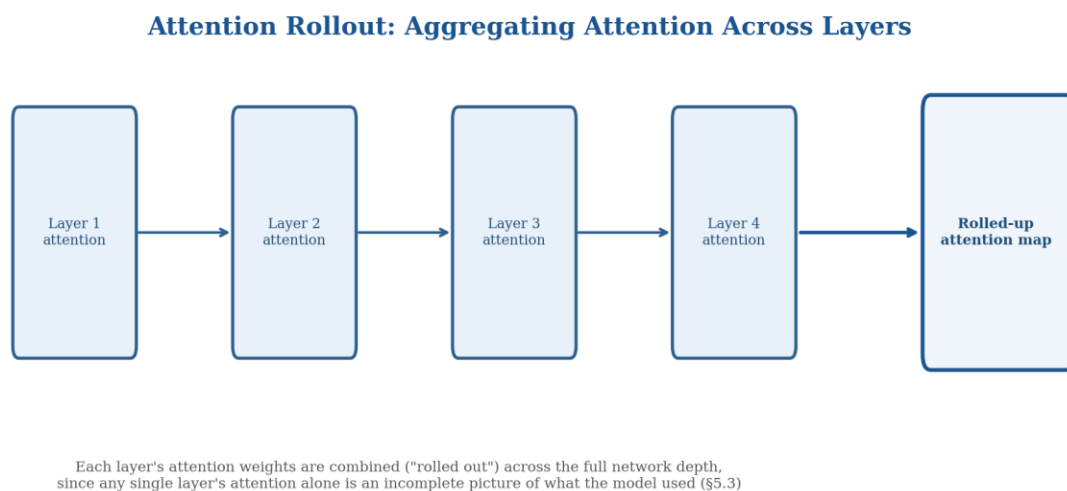


Figure 4. Attention rollout combines each layer's attention weights across the network's full depth into one aggregated map. A single layer's attention alone — the naive alternative — gives only a partial account of how information actually flowed from the input to the final prediction; rollout is a direct, practical response to that incompleteness.

5.4 A Practical Comparison Across All Three Method Families

Property	Gradient-Based (§3)	Perturbation-Based (§4)	Attention-Based (§5)
Requires model access?	Gradients only	None — black box	Attention weights only
Applicable architectures	Any differentiable model	Any model at all	Transformer-based only
Computational cost	Low	Higher	Very low — already computed

Property	Gradient-Based (§3)	Perturbation-Based (§4)	Attention-Based (§5)
Faithfulness confidence	Moderate, per §3.4	High, by construction	Debated, per §5.2

No single row of this table dominates the others — the same recurring pattern this compendium's every comparison section has established. A CNN-based medical classifier (PP13's Task A, developed in full in this day's assignment) has no attention weights to inspect at all, making Section 5's methods inapplicable regardless of their other merits, and Grad-CAM (Section 3) is the natural default. A transformer-based model offers attention maps essentially for free, but Section 5.2's faithfulness caveat means a careful practitioner should not treat them as a strictly superior substitute for an independent, gradient-based or perturbation-based check where the stakes justify the additional cost — precisely the stakes Section 6 argues medical imaging represents.

6 Medical Imaging as a High-Stakes CV Domain

With Sections 3 through 5's explainability toolkit fully developed, this section turns to the domain that makes that toolkit a genuine necessity rather than an optional add-on. Medical imaging is this module's natural closing application precisely because it draws together nearly every prior day's technical content into one setting, while adding a stakes dimension no earlier application in this compendium carried with comparable weight.

6.1 Why Medical Imaging Combines Nearly Every Prior Day's Content

Segmentation (Day 33) is directly required for organ and lesion delineation — precisely marking the boundary of a tumor, or the extent of an organ, for measurement or treatment planning. Classification (Day 29) is directly required for diagnostic categorization — pneumonia versus normal, malignant versus benign. Volumetric, 3D data (Day 36 §6) is the native format of CT and MRI, not an occasional special case. And explainability (Sections 3 through 5 of this day) is, as Section 1 argued, a clinical necessity rather than a nice-to-have. No earlier application in this compendium required this particular combination of capabilities simultaneously, which is exactly why medical imaging is positioned as this module's closing synthesis rather than simply one more application among several.

6.2 Regulatory and Trust Context

A clinician deciding whether to act on an AI model's flagged finding needs to know why the model flagged it before incorporating that finding into a clinical decision — not out of mere curiosity, but because the clinician bears the ultimate responsibility for the patient's care, and cannot responsibly defer that judgment to an opaque system whose reasoning cannot be inspected. This is a fundamentally different relationship between a human and a model's output than any earlier application in this compendium required: Day 31's retail inventory system does not need to justify its count to a store manager's satisfaction before restocking occurs, but a diagnostic imaging system's flagged finding sits within a clinical decision-making process

where the human retains, and must retain, final judgment — and exercising that judgment responsibly requires the kind of insight into model reasoning Sections 3 through 5 developed.

6.3 The Stakes of Errors in This Domain

It is worth being explicit about how the cost of an error in medical imaging differs from every earlier application this compendium covered — not merely in degree, but sometimes in kind. A false negative (a missed pathology) can result in delayed treatment and a worse patient outcome. A false positive (a healthy patient flagged as potentially unwell) can result in patient anxiety, unnecessary follow-up procedures, and their own associated risks and costs. Section 9.1 develops this asymmetry in full quantitative detail — including why the two error types are not simply interchangeable and cannot be minimized by the same strategy — but the qualitative point is worth establishing here first: unlike Day 31's detection latency budget (measured in milliseconds) or Day 34's generative sample quality (measured in FID score), an error in this domain is measured, ultimately, in terms of patient welfare — a categorically different kind of stake than this compendium has addressed directly until now.

Callout Box: *This day's stakes framing is not intended to suggest that medical imaging AI is uniquely dangerous or that earlier applications in this compendium were somehow careless by comparison — Day 26 §9's ethics discussion, Day 31 §9.4's surveillance concerns, and Day 34 §10.2's synthetic-media concerns all took their own respective domains' risks seriously throughout. The point is narrower and more specific: medical imaging is the domain where explainability (Sections 3-5) stops being merely a useful debugging tool, of the kind Day 29 §9.3 first demonstrated, and becomes a necessary component of the deployed system itself — a distinction Section 9.3 develops directly in the context of regulatory approval.*

7 Medical Image Modalities & Data Characteristics

Before developing the architectures Section 8 builds for this domain, it is necessary to understand the data itself — medical imaging spans several genuinely distinct modalities, each with its own dimensionality, intensity conventions, and practical data constraints, and a single generic "medical image" mental model would badly under-serve the real variety this section develops.

7.1 A Survey of Modalities

X-ray produces a single 2D projection image — light passing through tissue is absorbed differentially by different tissue densities, producing a 2D grayscale image with no depth information at all, the same fundamental single-projection limitation Day 36 §1 developed in the abstract, now the literal acquisition physics of a specific, extremely common modality. Ultrasound produces a 2D (or live video) image via reflected sound waves, with real-time acquisition and a quality that depends substantially on operator skill and probe positioning — a data-quality variability with no comparably direct analogue among this compendium's earlier standard datasets. Histopathology produces extremely high-resolution 2D images of stained tissue slides — commonly gigapixel-scale, orders of magnitude larger than any image this

compendium has otherwise processed directly, requiring specialized tiling strategies structurally similar to Day 33 §7.1's satellite-imagery tiling discussion. CT (computed tomography) and MRI (magnetic resonance imaging) both produce genuinely volumetric, 3D data — a full 3D voxel grid of intensity values, not a 2D image at all, directly instantiating the voxel-grid representation Day 36 §6.1 developed in the abstract as this domain's literal native data format, per Day 36 §9.4's forward reference to this day.

Medical Imaging Modalities: Dimensionality and Data Characteristics

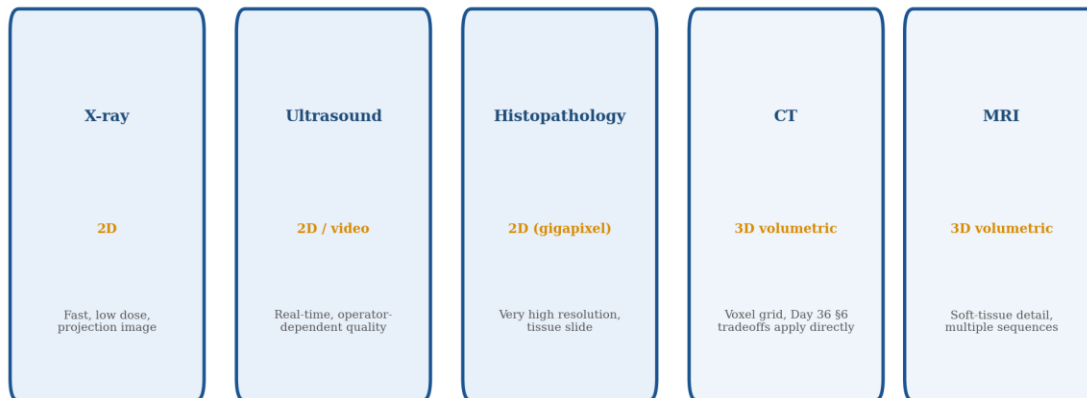


Figure 5. Medical imaging modalities span a genuine range of dimensionality and data characteristics. X-ray, ultrasound, and histopathology are 2D (or 2D-plus-time); CT and MRI are natively 3D volumetric data, meaning Day 36 §6's voxel-grid memory-versus-resolution tradeoff applies directly and with real clinical weight, not merely as an abstract architectural concern.

7.2 Grayscale, Intensity Conventions, and Resolution

Most medical imaging modalities are grayscale rather than color — X-ray, CT, and MRI all encode a single physical measurement (density, absorption, or tissue relaxation properties) per voxel rather than the three-channel RGB signal Day 26 §2.2 developed for natural photography, meaning color-based preprocessing and augmentation techniques from Day 27 §3.3 are largely inapplicable here, exactly as Day 32 §8.4's video preprocessing discussion flagged for a structurally similar reason. CT specifically uses a standardized intensity scale (Hounsfield units) with fixed physical meaning across different scanners and institutions, and a preprocessing step called windowing — clipping the full intensity range to a diagnostically relevant sub-range before display or model input — is a modality-specific preprocessing step with no equivalent among the natural-image pipelines Day 27 developed.

7.3 Data Scarcity and Privacy Constraints

Medical imaging datasets are characteristically far smaller than the natural-image datasets this compendium has otherwise relied on — ImageNet's 1.2 million images (Day 26 §6.3) has no comparably-sized medical imaging counterpart, both because acquiring and expertly annotating medical images is expensive and slow,

and because patient privacy regulations meaningfully restrict how medical imaging data can be collected, shared, and combined across institutions in the first place. This combination of factors is why transfer learning (Section 8.1) and the self-supervised pretraining strategies previewed in Section 10.2 are not merely convenient but close to essential in this domain, considerably more so than the classification and detection tasks earlier in this compendium where reasonably large labeled datasets were often assumed available.

7.4 Class Imbalance as a Common Challenge

Rare pathologies are, definitionally, rare — a diagnostic dataset drawn from real clinical cases will typically contain far more examples of a healthy or normal finding than of any specific rare condition, the same class-imbalance concern Day 33 §7.1's pixel-accuracy discussion raised for segmentation (where a small, rare structure can be nearly invisible to an aggregate metric), now recurring for classification specifically. A classifier trained naively on such an imbalanced dataset can achieve deceptively high aggregate accuracy simply by predicting the majority class most of the time — echoing Day 33 §7.1's own worked example of a model that "never detects the pedestrian" and still posts a high pixel-accuracy score — which is precisely why Section 9.1 develops sensitivity and specificity, rather than raw accuracy, as this domain's standard evaluation metrics.

8 Architectures for Medical Image Analysis

This section develops the direct payoff of two forward references made earlier in this compendium: Day 29 §7.5's transfer-learning workflow, and Day 33 §4 and §8's U-Net forward reference to this exact day. PP13's two-part structure — a classification task and a segmentation task, developed side by side in this section precisely because the assignment requires building one of each — maps directly onto the two architectural paths this section covers.

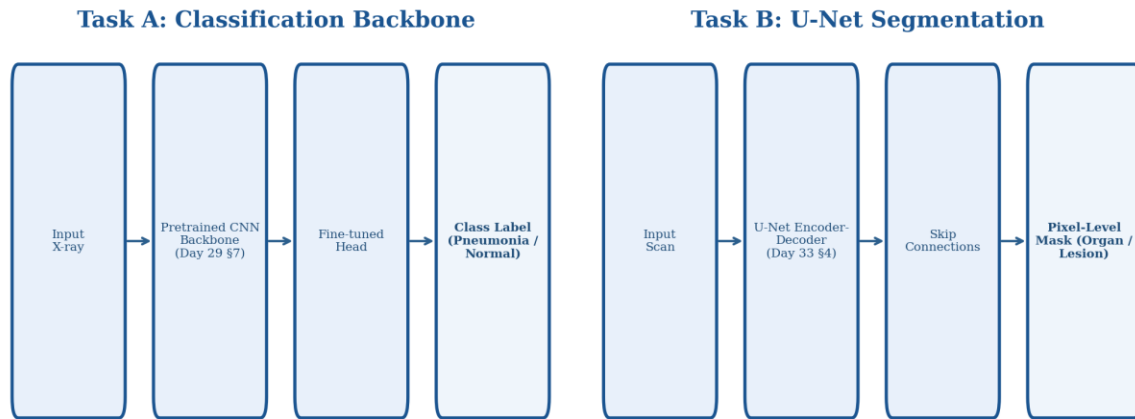


Figure 6. The two architectural paths PPI3 requires side by side: Task A (left) fine-tunes a pretrained CNN classification backbone, echoing Day 29 §7's transfer-learning workflow directly. Task B (right) uses a U-Net-style encoder-decoder with skip connections, the direct payoff of Day 33 §4's forward reference to this day.

8.1 The Classification-Backbone Path

Task A's classification path follows exactly the transfer-learning workflow Day 29 §7.5 developed: start from an ImageNet-pretrained backbone (any of Day 30's architecture family — a ResNet-50 remains a reasonable, well-supported default per Day 30 §4.3's own discussion of why it persists as the field's most common choice), replace the final classification head with one matching the target diagnostic categories, and fine-tune on the target medical dataset. This is directly comparable in structure to Day 26's PP12 face-mask classification project, now applied to a diagnostic imaging category set instead.

8.2 The Domain Gap: ImageNet Pretraining on Medical Images

A specific complication is worth developing carefully here, since it is easy to underestimate. ImageNet pretraining teaches a backbone to recognize the visual statistics of natural, everyday photographs — color photographs of common objects, animals, and scenes, captured under ordinary lighting. A chest X-ray is grayscale, has an entirely different intensity distribution, and contains visual structure (rib shadows, lung fields, soft-tissue density gradations) with no direct analogue among ImageNet's object categories. This domain gap means an ImageNet-pretrained backbone's early-layer features (edge and texture detectors, per Day 29 §9.1) transfer reasonably well — edges and gradients are a fairly universal low-level visual primitive — but its later, more class-specific layers transfer considerably less directly, and fine-tuning needs to adapt more of the network's later layers than a same-domain transfer-learning scenario would typically require, a consideration worth budgeting extra training time and a closer eye on validation performance for.

8.3 The U-Net Segmentation Path

Task B's segmentation path is the direct payoff of Day 33 §4.3's own explicit statement that U-Net "was originally developed for biomedical image segmentation" and Day 33 §8's forward reference naming this exact day. U-Net's skip-connection design — carrying fine spatial detail from encoder to matching decoder resolution, developed in full in Day 33 §4.2 — is particularly well suited to this domain's fine-detail-preservation needs: a tumor's precise boundary, or a small anatomical structure's exact extent, is often the entire clinically relevant content of a segmentation output, and U-Net's architecture was specifically designed to preserve exactly this kind of fine boundary detail that a naive, skip-connection-free encoder-decoder (Day 33 §3) would blur.

8.4 3D CNN and Volumetric Architectures for CT/MRI

For genuinely volumetric modalities — CT and MRI, per Section 7.1 — the 2D architectures developed above are extended directly into 3D, using 3D convolutions in place of 2D convolutions throughout, the same architectural extension Day 32 §8.2 introduced for 3D CNN-based activity recognition, now applied to a single volumetric scan rather than a video clip's temporal dimension. This carries a direct, non-trivial compute cost: a 3D convolutional kernel has meaningfully more parameters and requires meaningfully more computation than its 2D counterpart at the same spatial kernel size, echoing Day 36 §6.2's voxel-grid memory-cost discussion, now recurring as a computational cost during training and inference rather than merely a storage cost.

8.5 Dice Coefficient: The Standard Segmentation Metric Here

Day 33 §7.3 introduced the Dice coefficient and specifically flagged medical imaging as the domain where it is most commonly the reported metric of choice, ahead of raw pixel accuracy or even standard IoU — a direct consequence of Section 7.4's class-imbalance discussion: a small tumor or lesion occupying a tiny fraction of a scan needs a metric that scores partial, near-miss overlap more generously than IoU does, exactly the property Day 33 §7.3 identified as Dice's defining characteristic relative to IoU. PP13's Task B evaluation, developed in full in this day's assignment brief, uses Dice directly for this reason.

9 Evaluation, Regulation & Clinical Deployment

9.1 Sensitivity and Specificity: Why Accuracy Is Insufficient

Section 7.4 already flagged why raw accuracy is a poor primary metric under the class imbalance typical of real diagnostic datasets. Sensitivity (also called recall in this compendium's earlier terminology, Day 31 §3.1) measures the fraction of actually-positive cases the model correctly identifies — a model with low sensitivity misses real pathologies, producing false negatives. Specificity measures the fraction of actually-negative cases the model correctly identifies as negative — a model with low specificity over-flags healthy patients, producing false positives. Figure 7's left panel shows the two metrics' typical tradeoff as a model's

decision threshold varies: raising the threshold to catch more true positives (higher sensitivity) generally means also accepting more false positives (lower specificity), and vice versa — the same precision-recall tradeoff structure Day 31 §3.2 developed for detection, now recurring in a diagnostic classification context with a specific clinical framing attached to each axis.

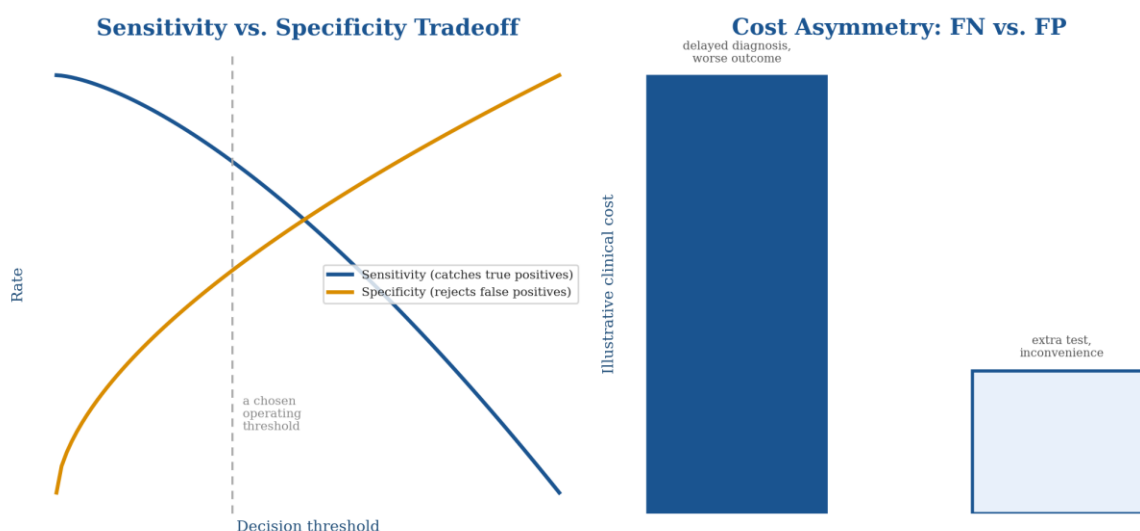


Figure 7. Left: sensitivity and specificity trade off against each other as the decision threshold varies, the same precision-recall structure Day 31 §3.2 developed, now in a clinical framing. Right: the illustrative cost asymmetry between a false negative (a missed pathology, generally the more clinically serious error) and a false positive (an unnecessary follow-up) — the reason threshold selection in this domain is a clinical judgment, not a purely statistical one.

9.2 The Cost Asymmetry Between False Negatives and False Positives

Section 6.3 introduced this asymmetry qualitatively; Figure 7's right panel makes the underlying reasoning concrete. A false negative in a cancer-screening context means a real pathology goes undetected and untreated until it is caught by some other means, potentially considerably later, with a correspondingly worse prognosis. A false positive means a healthy patient undergoes an unnecessary follow-up procedure — inconvenient, sometimes itself carrying a small independent risk (a biopsy, for instance), but categorically less severe than a missed diagnosis in most (though not all) diagnostic contexts. This asymmetry is precisely why medical diagnostic systems are frequently tuned to favor higher sensitivity at some real cost to specificity — deliberately accepting more false positives in exchange for fewer missed true positives — a deliberate, clinically-motivated departure from a threshold that would simply maximize raw accuracy, which is itself further evidence for Section 7.4's claim that accuracy alone is an inadequate metric in this domain.

9.3 Regulatory Context, at a Conceptual Level

Medical imaging AI systems intended for real clinical use are, in most jurisdictions, subject to formal regulatory review before deployment — in the United States, this generally falls under FDA review processes for medical devices and software; in the European Union, under CE marking requirements for

medical devices. This is worth understanding at a conceptual level only — this section is not legal advice, and the specific regulatory pathway a given system needs to follow depends on jurisdiction-specific detail well beyond this day's scope. The point worth taking away conceptually is this: explainability (Sections 3 through 5) is not merely a debugging convenience in this regulatory context, but frequently a direct component of what regulators and clinical reviewers expect a submission to demonstrate — a system whose predictions cannot be explained is, in practice, considerably harder to gain clinical trust and regulatory approval for, connecting Section 6.2's trust discussion directly to a concrete institutional consequence.

9.4 Dataset Bias and Generalization Across Hospitals and Populations

Day 26 §9.2 developed dataset bias as a general concern — a model can only be as representative as the data it was trained on — and this section extends that concern into its highest-stakes form. A medical imaging model trained on data from one hospital, using one specific scanner manufacturer's equipment, calibrated to one particular patient population's demographics, can fail to generalize when deployed at a different hospital with different equipment and a different patient population — not a hypothetical concern, but a well-documented, recurring failure pattern in real deployed medical imaging systems. This is Day 26 §9.2's Gender Shades-style demographic fairness concern, combined with an equipment- and institution-specific generalization concern with no direct analogue among this compendium's earlier, more visually homogeneous datasets (a webcam-captured face mask dataset, for instance, does not vary nearly as systematically by acquisition hardware as a CT scan does across different scanner manufacturers and imaging protocols).

Callout Box: *A documented, illustrative pattern worth internalizing directly: a diagnostic model trained and validated at a single academic medical center, achieving excellent performance there, subsequently showed a measurable performance drop when evaluated on imaging data from a different hospital using different scanner hardware — despite both hospitals imaging the same underlying clinical condition. The model had, in effect, partly learned to recognize the specific scanner's characteristic imaging signature alongside (or, in the worst documented cases, instead of) the actual pathology — a direct structural echo of Section 1.1's wolves-versus-huskies spurious-correlation case, now recurring with real patients rather than a benchmark dataset's photographs. Multi-institution validation, across genuinely varied equipment and populations, is the standard practical response — and explainability methods (Sections 3-5) are frequently what first reveals this kind of hardware-specific shortcut, exactly as Grad-CAM revealed the snow-background shortcut in Section 1.1's original case.*

10 Module Synthesis & Research Frontiers

This is the final day of the Computer Vision Fundamentals module, and this section is deliberately more retrospective than any earlier day's closing section — worth pausing on the full arc before turning to the remaining research frontiers and the module's forward connection.

10.1 The Module's Arc, Retraced

Day 26 opened the module by establishing the foundational task taxonomy and the classical computer vision pipeline, framing vision as an ill-posed inverse problem — a framing this exact day's Section 1 has now applied, in its highest-stakes form, to the question of trusting a model's diagnostic output. Day 27 developed image preprocessing as the load-bearing, easy-to-get-silently-wrong step between raw data and a trainable model — a lesson this day's Section 7.2 revisited directly for medical imaging's modality-specific intensity conventions. Day 28 developed classical feature detection and matching in full, providing the RANSAC-based geometric estimation machinery Day 36 later extended into 3D. Day 29 introduced CNNs from first principles, including, in §9, the Grad-CAM technique this exact day's Section 3 developed in full. Day 30 surveyed the decade of architectural evolution from VGG through Vision Transformers, providing the backbone-selection framework this day's Section 8.1 applied directly to a medical imaging classification task. Days 31 through 33 developed detection, tracking, and segmentation — with Day 33's U-Net receiving, in this exact day's Section 8.3, the dedicated biomedical-imaging treatment its own text explicitly deferred. Days 34 and 35 developed the module's generative arc, GANs and diffusion models, turning the discriminative-task framing of every earlier day around entirely. Day 36 added the third spatial dimension and temporal video processing, providing the volumetric representation concepts this day's Section 7.1 applied directly to CT and MRI. And this day closes the arc by bringing nearly every one of those threads — architecture choice, evaluation methodology, 3D representation, and now explainability — to bear on a single, unified, high-stakes application domain.

It is worth naming explicitly what makes this closing genuinely a synthesis rather than merely a list of callbacks: medical imaging did not require any fundamentally new architectural idea beyond what Days 26 through 36 already developed. Every technical building block — transfer learning, U-Net, 3D convolution, Dice coefficient, gradient-based explanation — was already available. What this day added was the domain-specific judgment needed to combine them responsibly: knowing that accuracy alone is insufficient (§9.1), that a model's high performance can hide a spurious shortcut (§1.1, §9.4), and that a technically correct model is not, by itself, a responsibly deployed one (§9.3) without the explainability component this day spent its first half developing.

10.2 Current Research: Foundation Models for Medical Imaging

Section 7.3 identified data scarcity as a defining constraint of this domain — far smaller labeled datasets than the natural-image benchmarks this compendium otherwise relied on. Foundation models, previewed across this compendium in Day 26 §10.1 (CLIP, DINO, SAM) and Day 30 §10.3 (large-scale pretrained backbones), offer a direct response: a model pretrained at very large scale on broad, often non-medical data, then adapted to a specific medical imaging task with comparatively little task-specific labeled data — the same transfer-learning logic Section 8.1 developed, extended to a considerably larger and more capable starting checkpoint. Self-supervised pretraining specifically — learning useful representations from unlabeled medical images directly, with no diagnostic labels required at all, echoing Day 30 §10.2's DINO and MAE discussion — is particularly well matched to this domain's specific data economics: unlabeled

medical images (routine clinical scans, for instance) are often considerably more abundant than expertly-labeled diagnostic datasets, the same labeled-versus-unlabeled abundance asymmetry Day 30 §10.2 identified as self-supervised pretraining's core motivating scenario in general.

10.3 Multimodal Medical AI

A growing research direction combines medical imaging with other clinical data modalities — a patient's clinical notes, laboratory results, and prior medical history — within a single model, echoing this compendium's broader multimodal thread (Day 26 §10.1's CLIP, Day 33 §8.2's SAM) but now combining genuinely different data types (images and structured or free-text clinical data) rather than images and natural-language category descriptions alone. The motivating clinical intuition is direct: a radiologist reading a scan does not read it in isolation, but in the context of a patient's full clinical picture, and a model with access to that same broader context has, in principle, a more complete evidentiary basis for its prediction than an image-only model can offer.

10.4 Forward References

This day closes the Computer Vision Fundamentals module, but the broader research programme continues immediately into a new, closely related module: Vision Transformers & Vision-Language Models, beginning with Day 38. That module picks up directly from threads this compendium has developed piecemeal since Day 30 §7 — Vision Transformers were introduced there as one architectural option among several; Day 38 onward gives them the full, dedicated treatment, including the inductive-bias data-hunger argument Day 30 §7.2 only sketched, and extends eventually into the vision-language models (Day 40 onward) that combine visual and textual understanding within a single system — a natural continuation of this exact day's multimodal medical AI discussion in Section 10.3, now generalized well beyond the medical domain.

10.5 Closing Perspective

Day 26 opened by observing that vision is a genuinely hard computational problem — an ill-posed inverse problem, requiring priors and assumptions to resolve. Eleven days and thirty-six days of calendar time later, this compendium has developed an extensive toolkit for building exactly those priors into a working system: classical features, learned representations, detection and segmentation architectures, generative models, 3D reconstruction, and now the explainability methods needed to trust a system's output in a setting where trust genuinely matters. Day 26 §9's opening ethics discussion asked whether a system's high accuracy could be trusted at face value; this day's entire second half has been a detailed, technically grounded answer: not without independent verification, and Sections 3 through 5 developed exactly the tools that verification requires. This is, deliberately, less a conclusion than a foundation — every technique in this compendium remains a starting point for the module beginning at Day 38, not a finished, closed body of knowledge.

11 Common Pitfalls: A Practical Debugging Checklist

This day combines explainability methods with a high-stakes application domain — a poor or misleading result can originate from either half, and the following checks are worth working through roughly in order.

- **Never trust a single explanation method's output in isolation.** Per §2.3 and §5.2's faithfulness caveats, a Grad-CAM heatmap or attention map is evidence to weigh, not a certified account of model reasoning — cross-check a surprising or high-stakes result with a second, structurally different method (a perturbation-based check against a gradient-based result, for instance) before drawing a firm conclusion.
- **Check per-class sensitivity and specificity separately, not only an aggregate accuracy figure.** Per §7.4 and §9.1, aggregate accuracy can hide poor performance on the specific rare-pathology class that matters most clinically.
- **Verify a classifier's high confidence is grounded in clinically plausible image regions, not a spurious shortcut.** Per §1.1 and §9.4's documented cases, apply Grad-CAM (§3) or occlusion sensitivity (§4.1) to a sample of confident predictions and check that the highlighted regions correspond to genuine anatomical or pathological structure, not an incidental artifact (scanner-specific markings, positioning equipment, or image borders).
- **Distinguish a domain-gap fine-tuning issue from a genuine architecture or data problem.** Per §8.2, an ImageNet-pretrained backbone's later layers transfer less directly to grayscale medical images than its early layers do — a plateauing fine-tuning loss may call for adapting more layers or a lower learning rate before assuming a deeper problem.
- **Confirm mask/label interpolation and evaluation are handled correctly for segmentation.** Per Day 33 §2.2's nearest-neighbor rule and §7.3's Dice-versus-IoU distinction, verify the correct metric and interpolation method are being used for Task B's evaluation specifically.
- **Check for class imbalance in your specific training split, not only the source dataset's overall description.** Per §7.4, an unlucky train/validation split can produce a locally more (or less) imbalanced training set than the full dataset's documented statistics suggest — verify directly rather than assuming.

A Note on Reproducibility

Explainability methods introduce their own sources of run-to-run variation beyond the standard training randomness this compendium has flagged throughout (weight initialization, augmentation, per Day 29 §8.2's reproducibility note). LIME's random sampling of perturbed superpixel combinations (§4.2) and SHAP's approximation strategies for large feature sets (§4.3) both involve randomness that can produce slightly different attributions across repeated runs on the identical input and model — worth fixing

explicitly, and worth checking for stability across a few repeated runs, before treating any single explanation as definitive for a high-stakes case.

A Note on Comparing Against Published Benchmarks

Published medical imaging benchmark results are frequently produced on carefully curated, single-institution datasets under favorable acquisition conditions, and — per §9.4's documented cross-hospital generalization concern — a model's published benchmark performance is not a reliable predictor of its performance on a genuinely different institution's data. A PP13 submission's classification or segmentation metrics should be compared primarily against your own held-out validation split under clearly documented conditions, not directly against a published leaderboard figure, exactly the same standard this compendium's every other graded assignment (Day 31 §8.3, Day 32, Day 34 §9.5) has applied to its own internal comparisons.

Glossary of Terms

This day introduced a substantial amount of specialized explainability and clinical vocabulary. The table below collects the most important terms in one place, each with a pointer back to the section where it was developed in full.

Term	Definition
Interpretability	A model whose internal structure is transparent enough to understand reasoning directly (§2.1)
Explainability	A post-hoc technique producing a justification for an opaque model's decision (§2.1)
Saliency map	A gradient-based map showing which input pixels most affect a prediction (§3.1)
Grad-CAM	A gradient-weighted class activation mapping technique producing a coarse localization heatmap (§3.2)
Occlusion sensitivity	A perturbation-based method measuring confidence change as regions are masked (§4.1)
LIME	A perturbation-based method fitting a local, interpretable surrogate model (§4.2)
SHAP	A game-theoretic feature-attribution method based on Shapley values (§4.3)
Attention rollout	Aggregating attention weights across all transformer layers into one map (§5.3)
Faithfulness (of an explanation)	Whether an explanation genuinely reflects a model's actual reasoning (§2.3, §5.2)
Sensitivity	The fraction of true-positive cases a model correctly identifies; also called recall (§9.1)
Specificity	The fraction of true-negative cases a model correctly identifies (§9.1)
Windowing	Clipping a CT scan's intensity range to a diagnostically relevant sub-range (§7.2)

Term	Definition
Hounsfield units	The standardized CT intensity scale with fixed physical meaning (§7.2)
Domain gap	The mismatch between a pretrained backbone's training distribution and a new target domain (§8.2)
Dice coefficient	The standard medical segmentation metric, more forgiving of small-structure boundary error than IoU (§8.5)

Chapter Summary: Key Takeaways

- **Explainability is a clinical necessity in medical imaging, not a debugging convenience (§1)** — the wolves-versus-huskies spurious-correlation case is a direct warning for exactly what a confident, unexplained model can get wrong.
- **Interpretability and explainability are distinct (§2)** — modern CNNs and Vision Transformers are not inherently interpretable, which is why post-hoc explanation methods (§3–5) are necessary at all.
- **Grad-CAM trades spatial precision for semantic relevance (§3)**, using a single backward pass to produce a coarse but class-discriminative localization heatmap — cheap, and the standard first-choice explanation method.
- **Perturbation-based methods (§4) require no model internals at all**, at higher computational cost — the only option when gradient access is unavailable, such as a third-party API-only diagnostic model.
- **Attention weights are a built-in explanation signal for transformers (§5)**, but their faithfulness is an active, unsettled research debate — not a strictly superior substitute for gradient- or perturbation-based verification.
- **Medical imaging combines nearly every prior day's content** — classification (Day 29), segmentation (Day 33), volumetric data (Day 36) — while adding a stakes dimension (§6) no earlier application carried with comparable weight.
- **Sensitivity and specificity, not raw accuracy, are this domain's standard metrics (§9.1)**, reflecting a genuine cost asymmetry between false negatives and false positives that a single accuracy figure cannot represent.
- **This day closes the Computer Vision Fundamentals module by synthesis, not by introducing new architecture (§10.1)** — every technical building block was already available; what this day added was the domain judgment to combine them responsibly.

Assignment / Practical Project Brief

PP13: Medical Imaging AI

★ GRADED ASSIGNMENT

This is a graded assignment, carrying a formal rubric (Section D), and this module's capstone project. The core requirement: complete both a classification task and a segmentation task on medical imaging data, and apply an explainability method (Sections 3–5) to both — the deliverable must demonstrate this day's two halves, medical imaging and explainability, working together, not as two separate, unconnected exercises.

A Core Requirements

Task A — Classification. Build a 2D classification model on a modality such as chest X-ray (a binary pneumonia-versus-normal split, or a small multi-class category set), using the CNN transfer-learning workflow developed in Section 8.1 and, before it, Day 29 §7. This task is directly comparable in structure to Day 26's PP12 face-mask-detection project — a pretrained backbone, a replaced classification head, fine-tuned on the target dataset.

Task B — Segmentation. Build an organ- or lesion-segmentation model using a U-Net-style architecture (Section 8.3, and the full U-Net development in Day 33 §4), evaluated with the Dice coefficient (Section 8.5, and Day 33 §7.3's original development).

Required explainability component for BOTH tasks. Apply a Section 3–5 method to Task A's classifier — Grad-CAM (§3.2) is the natural default, following the exact worked walkthrough developed in §3.1. Apply an adapted method to Task B's segmenter — a direct Grad-CAM application does not transfer cleanly to a per-pixel segmentation output the way it does to a single-label classification score; a reasonable adaptation is computing feature attribution on the encoder for a specific predicted region of interest, or applying occlusion sensitivity (§4.1) at the region level rather than the whole-image level. Document whichever adaptation you choose and why it is an appropriate fit for a segmentation output specifically.

B Dataset

A publicly available, clearly licensed-for-educational-use dataset is required for each task, sized appropriately for Colab T4 constraints within a single working session.

Task	Dataset Option	Note
A — Classification	A public chest X-ray dataset (pneumonia/normal or a small multi-class set)	Widely available; watch for class imbalance per §7.4 — check your specific split

Task	Dataset Option	Note
B — Segmentation	A compact organ or lesion segmentation dataset (e.g. a lung-field or small-lesion set)	Prefer a dataset with an established train/val split for fair comparison

C Deliverables

- A trained classifier (Task A) and a trained segmenter (Task B), both fine-tuned from a pretrained starting point, not trained from random initialization.
- Explainability visualizations for both: Grad-CAM (or equivalent) heatmaps for a representative sample of Task A predictions, and your adapted method's output for a representative sample of Task B predictions.
- Standard classification metrics for Task A: sensitivity and specificity (§9.1), not accuracy alone — report both explicitly, at a stated decision threshold.
- Standard segmentation metrics for Task B: Dice coefficient and IoU (§8.5, Day 33 §7.2–§7.3).
- A brief written reflection connecting your explainability outputs directly to §9's clinical-trust discussion — does the highlighted region correspond to plausible clinical evidence, and what would you conclude if it did not?

D Grading Rubric

Rubric Component	Weight	What's Assessed
Task A implementation	20%	Correct transfer-learning fine-tuning; sensitivity/specificity reported at a stated threshold
Task B implementation	20%	Correct U-Net-style architecture; Dice/IoU correctly computed
Explainability component (both tasks)	30%	Appropriate method applied to each task; adaptation for segmentation is well-reasoned, not merely copied from Task A
Written reflection	15%	Genuine engagement with §9's clinical-trust framing, grounded in your own actual results
Code quality & write-up	15%	Clean, documented code; clear explanation of dataset, splits, and methodology

Note the weighting deliberately: the explainability component alone accounts for 30% of the grade, and combined with the written reflection, the explainability half of this assignment outweighs either individual modeling task. PP13 is a capstone specifically because it requires medical imaging and explainability

working together — a technically strong classifier or segmenter with a superficial or missing explainability component will not score well, regardless of raw predictive performance.

E Tips and Common Pitfalls

- **Check for class imbalance before training, not after.** Per §7.4, compute your training split's actual class distribution directly rather than assuming the source dataset's documented statistics apply unchanged to your specific split.
- **Watch for overfitting on a small medical dataset.** Per §7.3, medical datasets are characteristically smaller than natural-image benchmarks — monitor validation loss closely (Day 29 §11's checklist) and consider stronger regularization than a comparable natural-image task would need.
- **Budget extra fine-tuning attention for the ImageNet-to-medical domain gap.** Per §8.2, a plateauing Task A training curve may call for adapting more of the pretrained backbone's later layers, not just the final classification head.
- **Do not naively copy Task A's explainability method onto Task B unchanged.** Per §A's requirement, Grad-CAM's class-score-gradient formulation does not directly apply to a per-pixel segmentation output — document your adaptation explicitly.
- **Cross-check any surprising explainability result before reporting it as a finding.** Per §11's checklist, a single method's output is evidence to weigh, not a certified conclusion — spot-check a second method where a result seems clinically implausible.

F Submission Checklist

- Task A classifier fine-tuned from a pretrained backbone and evaluated with sensitivity and specificity
- Task B segmenter built with a U-Net-style architecture and evaluated with Dice coefficient and IoU
- Grad-CAM (or equivalent) applied to a representative sample of Task A predictions
- An adapted explainability method applied to a representative sample of Task B predictions, with the adaptation documented
- Written reflection connecting explainability outputs to §9's clinical-trust discussion, grounded in your own results
- Class imbalance in your actual training split checked and reported
- Code is documented, and a short README explains how to reproduce your results

End of Day 37 — final day of the Computer Vision Fundamentals module. Ready for gap review and expansion into full compendium.